

Assignment 1

ME Analysis 1, Fall 2026
Return in class on Sep. 16.
100 points

- 1) Prove that: If $f(x) = ax^3 + bx^2 + cx + d$ has two critical numbers, their average is the abscissa at the point of inflection. If there is just one critical number, it is the abscissa at the point of inflection.
- 2) If $y(x) = \frac{x+1}{x^2}$,
 - Plot this curve from $-10 \leq x \leq 10$ in Matlab, include plot in submission.
 - Show that $x = -2$ is a critical point, calculate the relative extremum here.
 - Show that $x = -3$ is an inflection point.
 - Discuss the asymptotes of this curve.
- 3) If $y(x) = 1 - \frac{x^2}{x-1}$,
 - Plot this curve from $-10 \leq x \leq 10$ in Matlab, include plot in submission.
 - Obtain vertical/horizontal/oblique asymptotes.
 - Discuss local maximum and minimum
 - Discuss concave and convex domains of the function.
- 4) A rectangular box (having edges, x , y and z) without lid has to be made from a material with surface area 16cm^2 . What is the maximum volume of that box?
- 5) The plane, $4x - 3y + 8z = 5$, and the cone, $z^2 = x^2 + y^2$, intersects along an ellipse. Find the highest and lowest points on this ellipse (Hint: Corresponds to the max. and min. of the function, $f(x, y, z) = z$).
- 6) Find the points on the sphere, $x^2 + y^2 + z^2 = 4$, that are closest to and farthest from the point, $(3, 1, -1)$.
- 7) Consider a power series, $f(x) = \sum_{n=0}^{\infty} T_n$, where, $T_n = a_n(x - x_0)^n$. It converges if $\left| \frac{T_{n+1}}{T_n} \right| < 1$. The corresponding value of $|x - x_0|$ is known as the radius of convergence. Show that the following series has a radius of convergence, $8/25$. Series is defined by: $T_n = \frac{(-5)^{2n}}{n^2 8^{n+3}} (x - 2)^n$.
- 8) Expand $f(x) = \sin(x)$ using Taylor series about $x = \pi/4$. Write the series in its compact summation notation (Hint: You will need two groups of summation terms).
- 9) Find Taylor series of $f(x) = x^2 + x + 1$ about:
 - (a) $a = 2$. Use the Taylor series expression to compute the function value at $x = 1$.
 - (b) $a = 5$. Use the Taylor series expression to compute the function value at $x = 1$.How do they compare? Why so?
- 10) Find Taylor series (include first five terms) of $f(x) = 1/\sqrt{x}$ about:
 - (a) $a = 2$. Use the Taylor series expression to compute the function value at $x = 1$.
 - (b) $a = 5$. Use the Taylor series expression to compute the function value at $x = 1$.How do they compare? Why so?